Siva G Nair

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Summary

Experienced AI Engineer with a proven ability to implement state-of-the-art solutions in computer vision, NLP, and predictive analytics. Developed and deployed deep learning models for industrial applications, including defect detection, assembly verification, and document automation. Good grasp of algorithms and data structures and a strong understanding of Python libraries such as Keras, TensorFlow, PyTorch, and OpenCV. A lifelong learner, keen to use new technologies to build purposeful AI apps.

Education

APJ Abdul Kalam Technological University, Thiruvananthapuram *June 2020 – March 2024*
Sree Buddha College of Engineering, Pattoor
Bachelor of Technology in Computer Science, Specialized in AI and ML

- CGPA: 6.48/10.0

Experience

AI Engineer *Kochi, Kerala*
Ombrulla, Kochi, Kerala *Dec 2024 – May 2025*

- Developed and deployed industrial AI Proofs of Concept (POCs) for computer vision and natural language processing (NLP), including large language model (LLM)-based document processing systems, achieving 90% accuracy in automated document classification; delivered on-site client demos for visual inspection solutions, resulting in 30% faster client feedback cycles.
- Engineered a Retrieval-Augmented Generation (RAG)-based NLP system for document question answering, tabular data interaction, and automated summarization, improving information retrieval efficiency by 40% and enabling real-time decision-making for enterprise clients.
- Collaborated on a vision inspection application, integrating camera systems with deep learning models (CNNs) using PyTorch and OpenCV, enabling local training and deployment of machine learning models with 95% defect detection accuracy.
- Optimized data pipelines for real-time processing of large-scale datasets (over 1M records), leveraging cloud platforms like AWS and Python libraries (Pandas, NumPy) to streamline model training and inference.

Data Science Intern *Kochi, Kerala*
Luminar Technolab, Kochi, Kerala *May 2024 – Nov 2024*

- Developed and optimized machine learning models for predictive analytics, achieving a 25% improvement in model accuracy for time-series forecasting using Scikit-learn, TensorFlow, and Keras.
- Conducted end-to-end data preprocessing, cleaning, and feature engineering on datasets exceeding 500K records, utilizing Pandas and NumPy to ensure high-quality input for model training.
- Created interactive data visualizations using Matplotlib, Seaborn, and Power BI, enabling stakeholders to identify key trends and reducing decision-making time by 20%.
- Implemented statistical analysis techniques, including hypothesis testing and regression analysis, to derive actionable insights from complex datasets, supporting business intelligence initiatives.

Projects

RAG-Based Email Automation Agent

- Developed a Retrieval-Augmented Generation (RAG) based system to automate email responses by extracting and analyzing content, including text and images, from emails and attachments.
- Implemented event detection to trigger notifications for key events mentioned in emails, improving response efficiency and user experience.

- Utilized EasyOCR and PyTorch for image processing and optical character recognition (OCR) to extract text from email attachments.
- Leveraged Ollama for local deployment of language models and Milvus DB for efficient vector storage and retrieval to power RAG-based contextual responses.

Botanic Vision (Rice Plant Disease Detection App) *Link*

- Developed a machine learning model to identify five major rice plant diseases with an accuracy of 93%.
- Tools Used: Python, Flask, TensorFlow

Text Emotion Detection *Link*

- Designed an NLP model to analyze and classify text into five emotional categories (e.g., joy, sadness, anger, fear, surprise) with an accuracy of 85%.
- Tools Used: Python, Streamlit, NLTK, Keras

Trash Detection *Link*

- Developed a convolutional neural network (CNN) to classify waste into categories such as plastic, metal, paper, organic waste, and trash.
- Tools Used: Python, Streamlit, OpenCV, Keras

Technical Skills

Languages: SQL, Python, HTML, JavaScript, PSQL

Data Visualization: Power BI, Matplotlib, Seaborn

Tools: Git, Visual Studio Code, PyCharm, Tesseract, EasyOCR, Ollama, Milvus

Hosting and Deployment: Cloudflare, Heroku, GitHub

Achievements

- Collaborated with peers as team captain, steering a four-member project toward successful completion, culminating in an impactful publication featured in IJERCSE's esteemed engineering journal.
- Earned a certificate from Tinkerhub for creating a machine learning model.